

Subject Code: 01CT0716

Subject Name: Mobile and Pervasive Computing

B. Tech. Year – IV (Semester VII)

Objective:

The objectives of this course are

- To understand the pervasive computing and its applications
- To introduce the concepts of mobile communication systems and pervasive computing.
- To understand the concepts of emerging wireless technologies.
- To be aware of pervasive computing practices.

Credits Earned: 05 Credits

Course Outcomes: After completion of this course, student will be able to:

1. Understand the fundamental concepts of pervasive computing.
2. Analyze the cellular network architecture and protocols.
3. Analyze various wireless network systems.
4. Design and implement pervasive application systems.

Pre-requisite of course:

Analog and digital communication, signals & systems, probability and statistics, Algorithm and data structures, and Object oriented programming

Teaching and Examination Scheme:

Teaching Scheme (Hours)			Credits	Theory Marks			Tutorial / Practical Marks		Total Marks
				E	I		V	T	
Theory	Tutorial	Practical		ESE	IA	CSE	Viva	Term Work	
04	00	02	05	50	30	20	25	25	150

Contents:

Unit	Topics	Hours
1	Overview of Wireless Systems: Infrastructure-based vs Ad-hoc, Wireless LANs, Cellular systems, Sensor networks, Bluetooth, WiFi, WiMAX, Zigbee, RFID	10
2	Medium Access: Link Adaptation, Routing Protocols	5
3	Mobility and Handoff Management: Link layer mobility mechanisms (location management protocols), Network layer mobility mechanisms (Macro and Micro mobility protocols), Handoff management protocols.	6
4	Cellular Networks: LTE and 5G overview, GSM network architecture, protocols, frequency allocation, Mobility management	10
5	Pervasive Computing: Principles, Characteristics, Pervasive devices, Smart sensors and actuators, Human-machine Interfaces, Biometrics	8
6	Context Aware Sensor Networks: Open protocols (SDP, Jini, SLP, UPnP), SyncML framework, Context aware mobile services, Context aware security.	4
7	Energy Efficiency, Energy harvesting, Localization (GPS/WiFi/GSM), Security	4
8	Recent Advances in Wearable devices	5
9	Body Area Networks (BAN)	4
Total Hours		56

Suggested Text books / Reference books:

1. I. STOJMENOVIC (2002), Handbook of Wireless Networks and Mobile and Pervasive Computing, Wiley.
2. S. LOKE (2006), Context-aware Pervasive Systems: Architectures for a New Breed of Applications, CRC Press.
3. A. OSSEIRAN, J.F. MONSERRAT, P. MARSCH, (Eds.), 5G Mobile and Wireless Communications Technology, Cambridge University Press, 2016
4. R. KAMAL (2008), Mobile and Pervasive Computing, Oxford University Press, 3rd Edition
5. L. MERK, M. NICLOUS (2006), Principles of Mobile and Pervasive Computing, Dreamtech Press, 2nd Edition.
6. G. AGGELOU (2004), Mobile Ad hoc Networks: From Wireless LANs to 4G Networks, McGraw-Hill Professional
7. J.J. DRAKE, Z. LANIER, C. MULLINER, P.O. FORA, S.A. RIDLEY, G. WICHESKI (2014), Android Hacker's Handbook, John Wiley, 1st Edition
8. Ashok K.Talukder and Roopa R.Yuvagal, "Mobile Computing", 2nd Edition, Tata McGraw Hill, 2010.
9. JochenBurkhardt, Horst Henn, Stefan Heper, Klaus Rindtorff and Thomas Schack,"Pervasive Computing Technology and Architecture of Mobile Internet Applications"Addison Wesley, 2002.

10. UweHansmann, L. Merk, M. Nicllous, T. Stober and U.Hansmann, "Pervasive Computing", Springer Verlag, 2003. Johcehn H.Schiller, "Mobile Communications", Addison-Wesley, 2003.

Suggested Theory distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve an effective teaching- learning process.

Distribution of Theory for course delivery and evaluation					
Remember	Understand	Apply	Analyze	Evaluate	Create
10%	20%	25%	20%	10%	15%

Suggested List of Experiments:

1. Introduction to Android Integrated Development Environment. Develop an application that displays "Hello World !!!" message.
2. Create a login Activity. It asks for "username" and "password" from the user. If the username and password are valid, it displays a Welcome message using new activity.
3. Develop a calculator android application.
4. Develop an application for multimedia processing.
5. Developing an application for data persistence.
6. Develop an application to establish http connection and toast notification.
7. Develop an application that auto sends received SMS.
8. Develop an application that identifies the Bluetooth devices in the wireless range.
9. Develop an application for text communication using Bluetooth.
10. Develop an application that prints the signal strength of Wi-Fi connection
11. Develop an application that marks the present location of the device on google map.
12. Develop an application that demonstrates use of any built-in sensor.

Supplementary Resources:

1. <https://nptel.ac.in/courses/106106147/>
2. https://www.coursera.org/videos/5g-training-qualcomm/hJkVd?query=mobile+computing&source=search&utm_campaign=programId%3ANNqDyxeeEeyRdxIhkRqYvw%3Bday%3A1632207600000%3Binvitation&utm_medium=email&utm_source=other
3. <https://www.coursera.org/learn/wireless-communications>